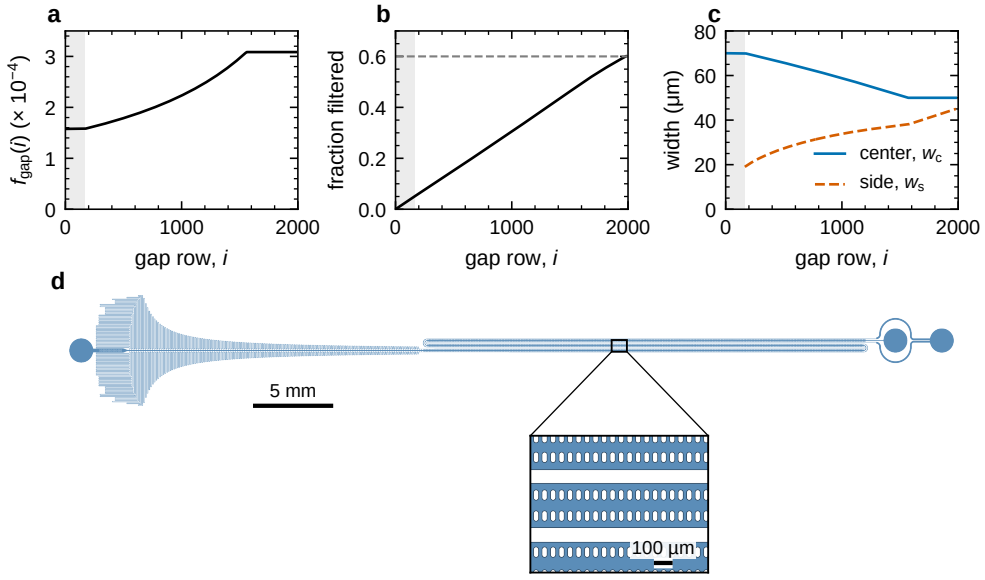
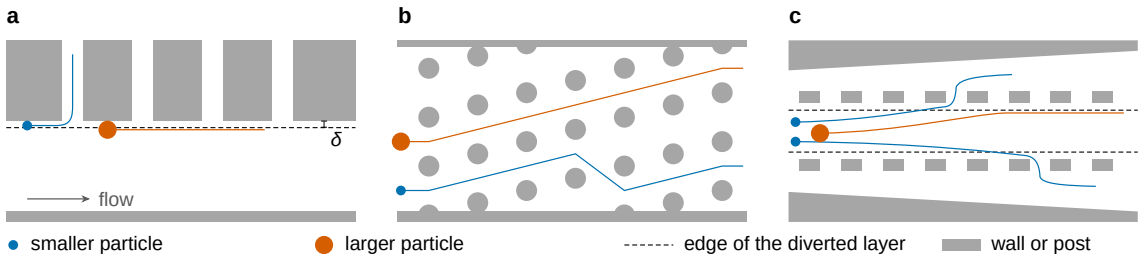
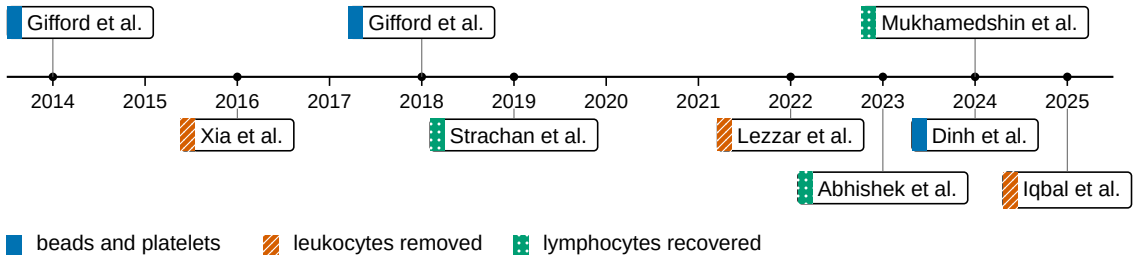
Dissertation figures, outline v1.2: proof sheet. Each figure is placed at its own size, at most the 6.0 in text width, with its register ID only. Captions, sources, permissions and alternative text are in registers/figures.json; the figures carry no captions, titles, notes or credit lines.


a
b
c
d
$f_{gap}(i)$ (× 10^{-4})
fraction filtered
width (μm)
gap row, i
gap row, i
gap row, i
center, w_c
side, w_s
5 mm
100 μm

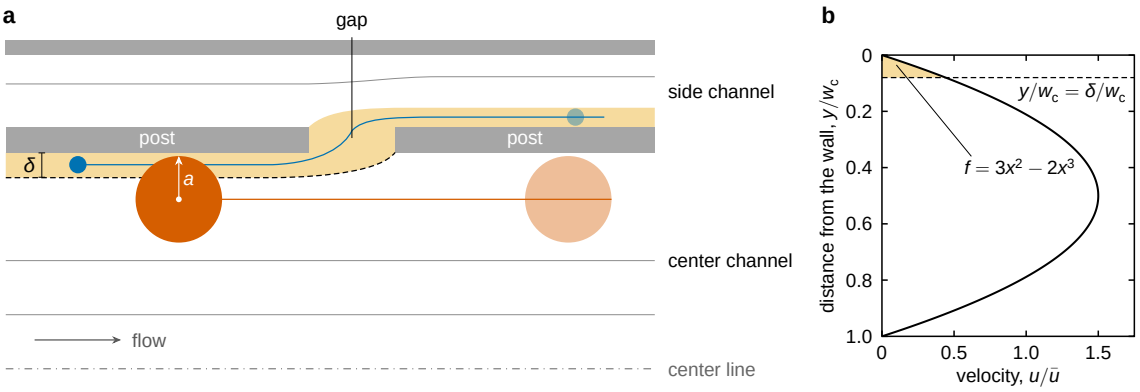
F1-1


a
b
c
δ
flow
smaller particle
larger particle
edge of the diverted layer
wall or post

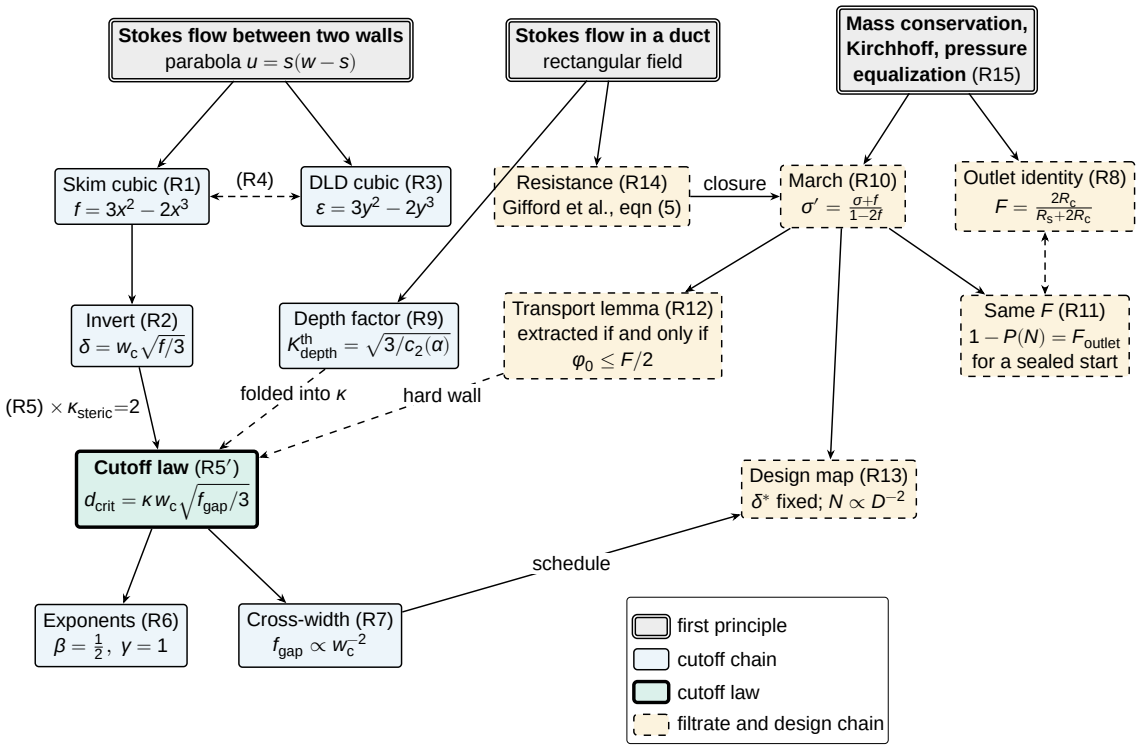
F2-1


Gifford et al.
Gifford et al.
Mukhamedshin et al.
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Xia et al.
Strachan et al.
Lezzar et al.
Dinh et al.
Abhishek et al.
Iqbal et al.
beads and platelets
leukocytes removed
lymphocytes recovered

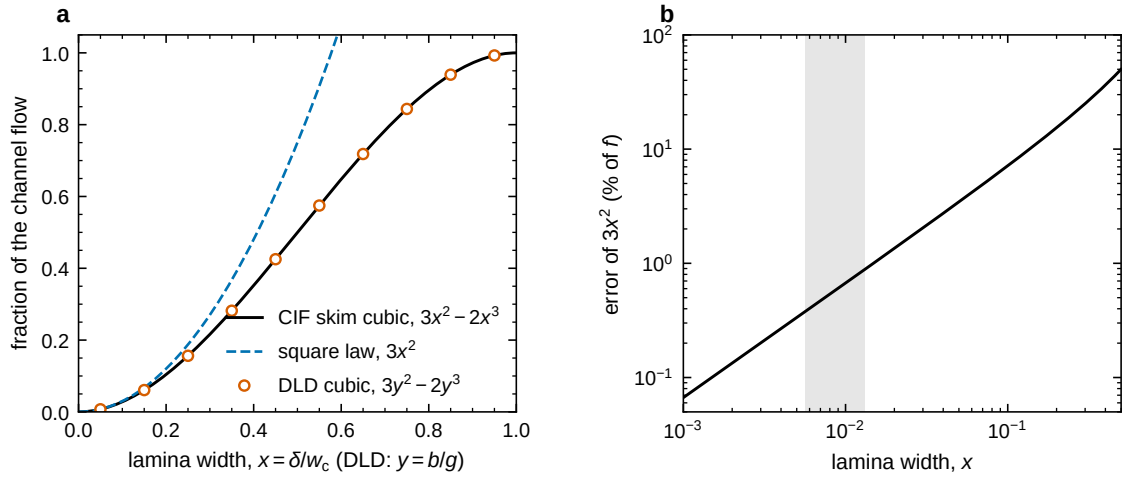
F2-2


a
gap
side channel
post
post
δ
a
center channel
flow
center line
b
0
0.2
0.4
0.6
0.8
1.0
distance from the wall, y/w_c
$y/w_c = \delta/w_c$
$f = 3x^2 - 2x^3$
0
0.5
1.0
1.5
velocity, $u/\bar{u}$

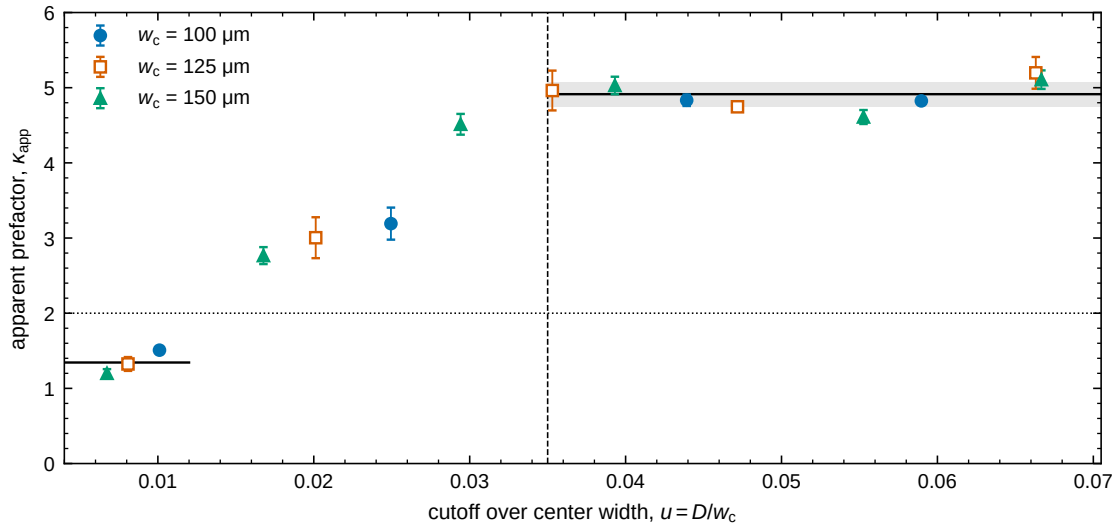
F3-1


Stokes flow between two walls
parabola $u = s(w - s)$
Stokes flow in a duct
rectangular field
Mass conservation, Kirchhoff, pressure equalization (R15)
Skim cubic (R1)
$f = 3x^2 - 2x^3$
(R4)
DLD cubic (R3)
$\varepsilon = 3y^2 - 2y^3$
Resistance (R14)
Gifford et al., eqn (5)
closure
March (R10)
$\sigma' = \frac{\sigma+f}{1-2f}$
Outlet identity (R8)
$F = \frac{2R_c}{R_s+2R_c}$
Invert (R2)
$\delta = w_c\sqrt{f/3}$
Depth factor (R9)
$K_{depth}^{th} = \sqrt{3/c_2(\alpha)}$
Transport lemma (R12)
extracted if and only if
$\varphi_0 \le F/2$
Same F (R11)
$1 - P(N) = F_{outlet}$
for a sealed start
(R5) $\times\, \kappa_{steric}$=2
folded into κ
hard wall
Cutoff law (R5′)
$d_{crit} = \kappa w_c\sqrt{f_{gap}/3}$
Design map (R13)
δ^* fixed; $N \propto D^{-2}$
schedule
Exponents (R6)
$\beta = \frac{1}{2},\ \gamma = 1$
Cross-width (R7)
$f_{gap} \propto w_c^{-2}$
first principle
cutoff chain
cutoff law
filtrate and design chain

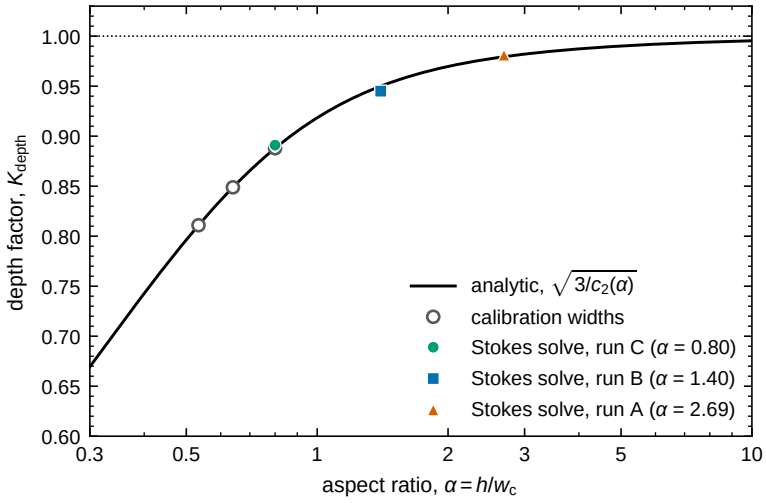
F3-2


a
b
fraction of the channel flow
lamina width, $x = \delta/w_c$ (DLD: $y = b/g$)
CIF skim cubic, $3x^2 - 2x^3$
square law, $3x^2$
DLD cubic, $3y^2 - 2y^3$
error of $3x^2$ (% of f)
lamina width, x

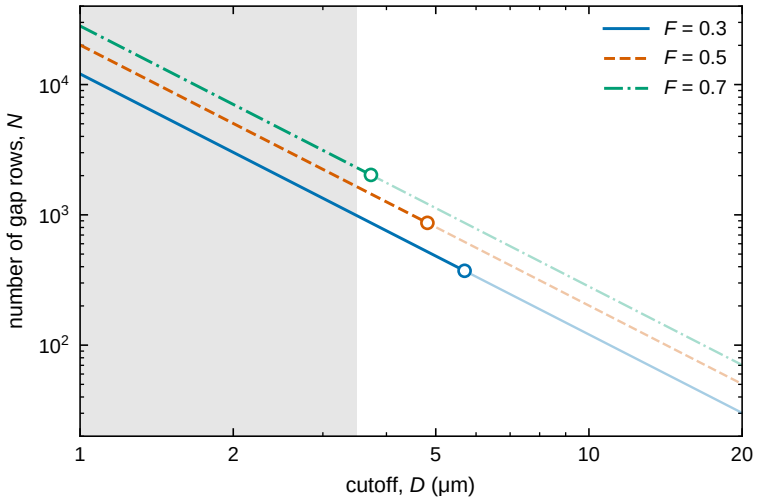
F3-3


w_c = 100 µm
w_c = 125 µm
w_c = 150 µm
apparent prefactor, κ_{app}
cutoff over center width, $u = D/w_c$
0
1
2
3
4
5
6
0.01
0.02
0.03
0.04
0.05
0.06
0.07

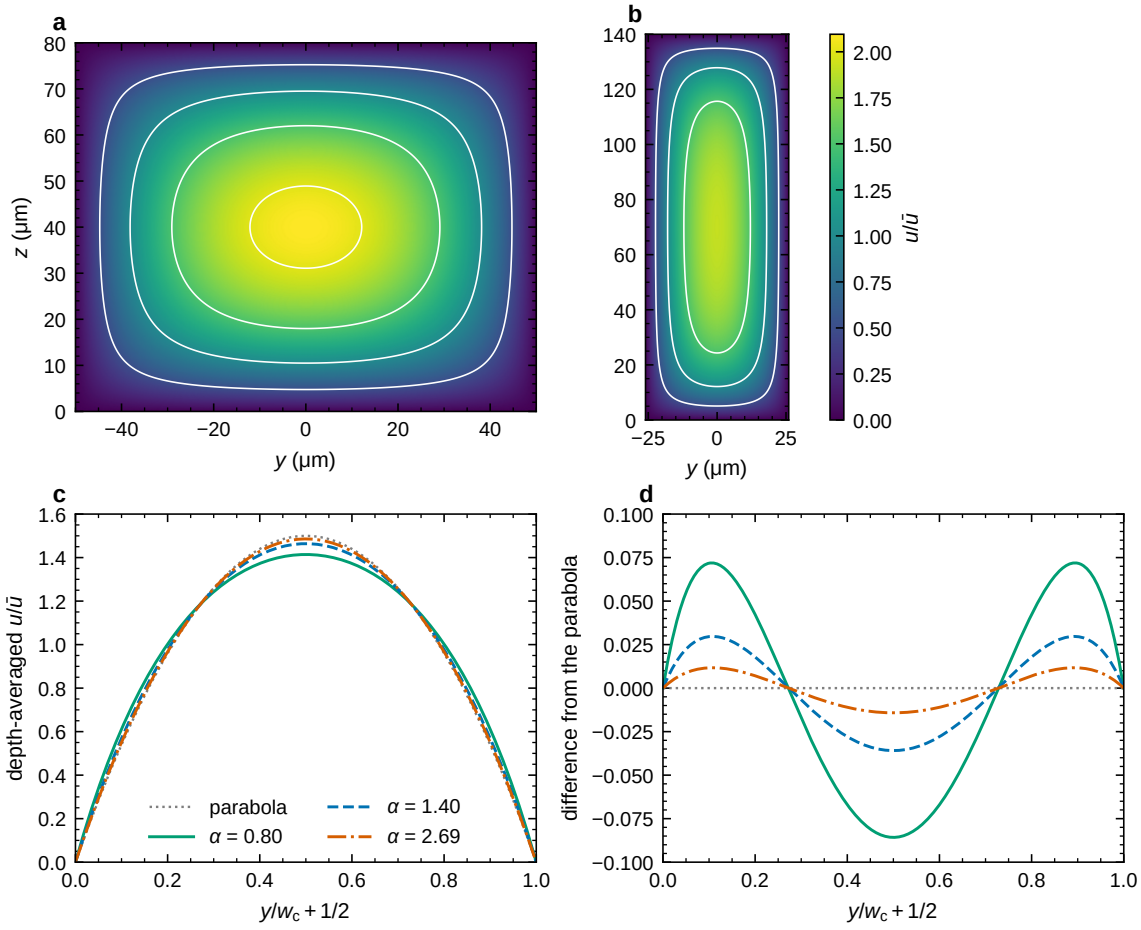
F3-4


1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
depth factor, K_{depth}
0.3
0.5
1
2
3
5
10
aspect ratio, $\alpha = h/w_c$
analytic, $\sqrt{3/c_2(\alpha)}$
calibration widths
Stokes solve, run C (α = 0.80)
Stokes solve, run B (α = 1.40)
Stokes solve, run A (α = 2.69)

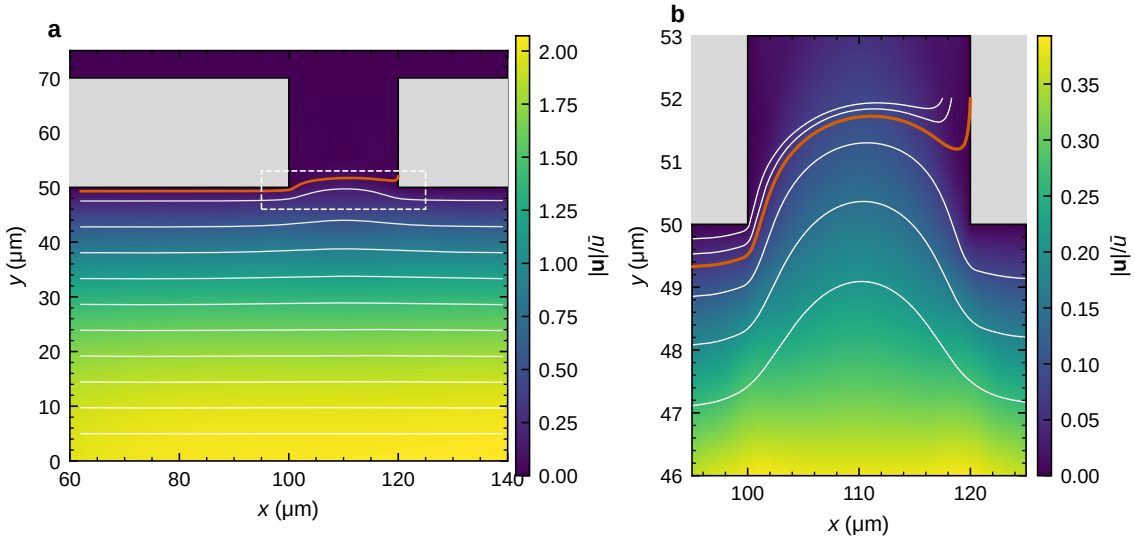
F3-5


F = 0.3
F = 0.5
F = 0.7
number of gap rows, N
10^4
10^3
10^2
1
2
5
10
20
cutoff, D (μm)

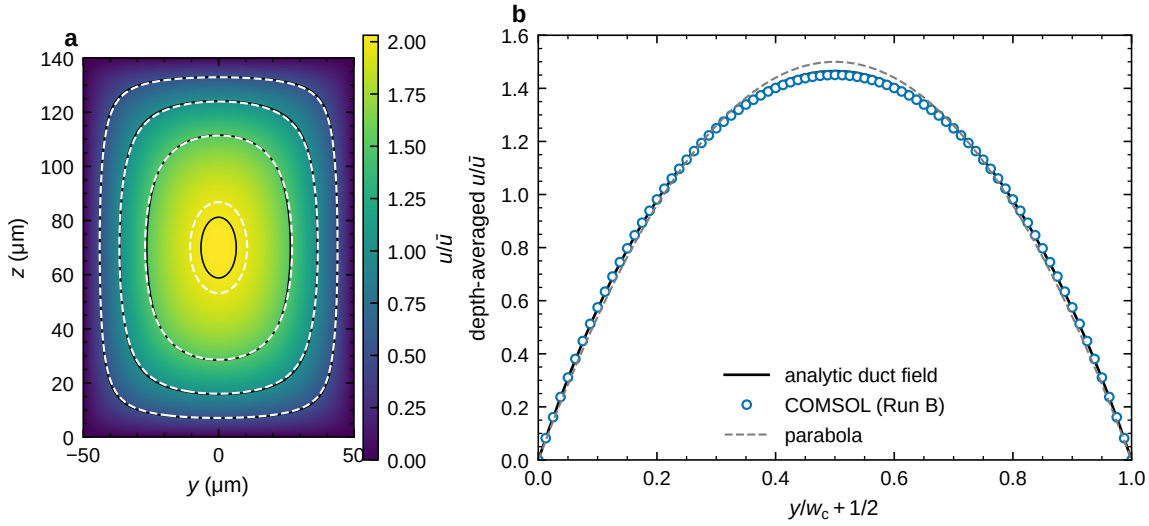
F3-6


a
b
c
d
z (μm)
y (μm)
u/ū
depth-averaged u/ū
difference from the parabola
y/w_c + 1/2
parabola
α = 0.80
α = 1.40
α = 2.69

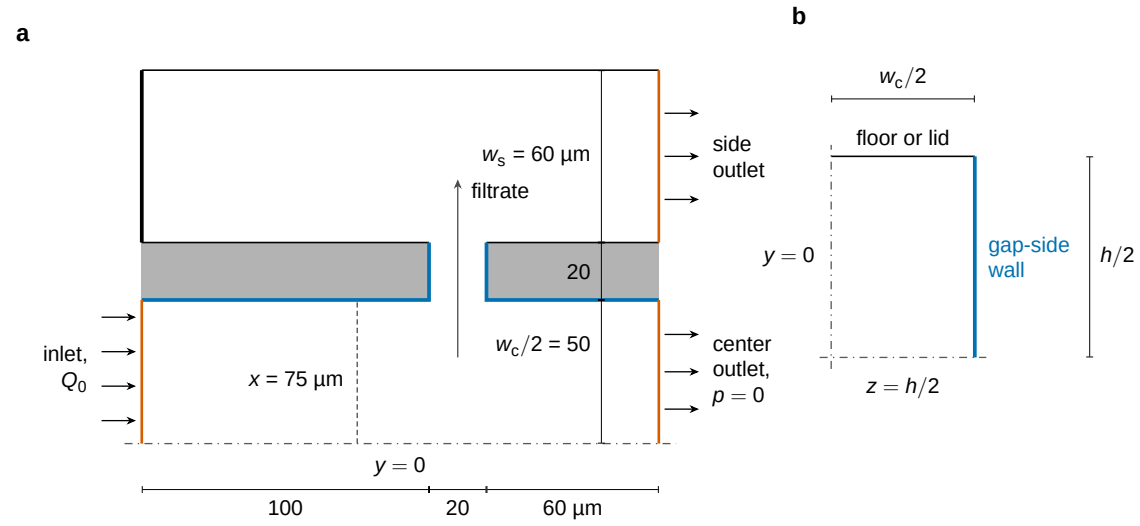
F3-7


a
b
y (μm)
x (μm)
|u|/ū
70
60
50
40
30
20
10
0
60
80
100
120
140
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
53
52
51
50
49
48
47
46
110
0.35
0.30
0.25
0.20
0.15
0.10
0.05

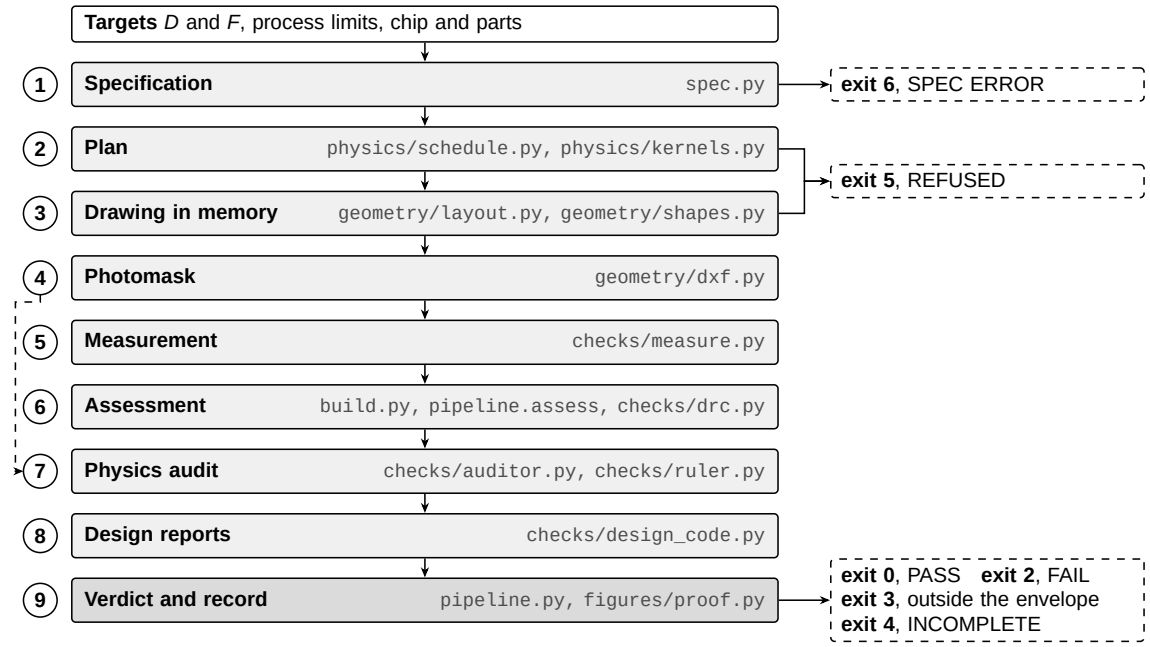
F4-1


a
b
z (μm)
y (μm)
u/ū
depth-averaged u/ū
y/w_c + 1/2
analytic duct field
COMSOL (Run B)
parabola

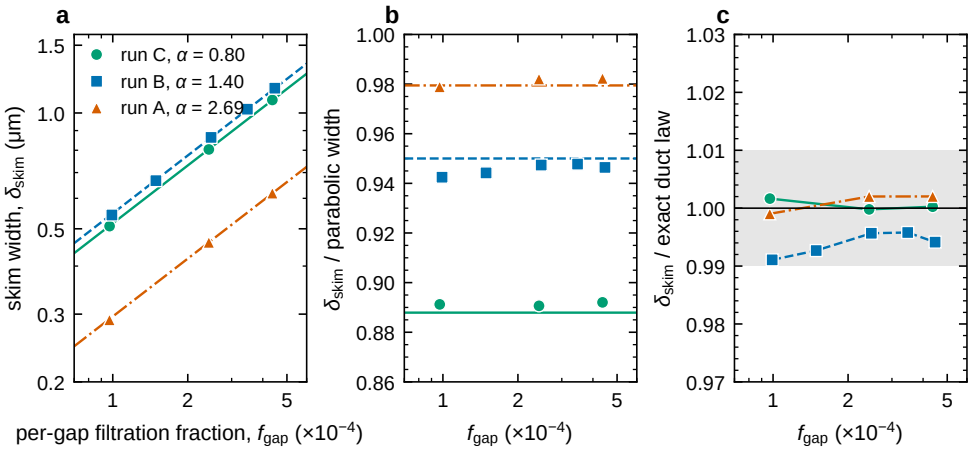
F4-2


a
b
w_s = 60 µm
filtrate
side
outlet
20
inlet,
Q_0
x = 75 µm
$w_c/2$ = 50
center
outlet,
$p = 0$
$y = 0$
100
20
60 µm
$w_c/2$
floor or lid
$y = 0$
gap-side
wall
$h/2$
$z = h/2$

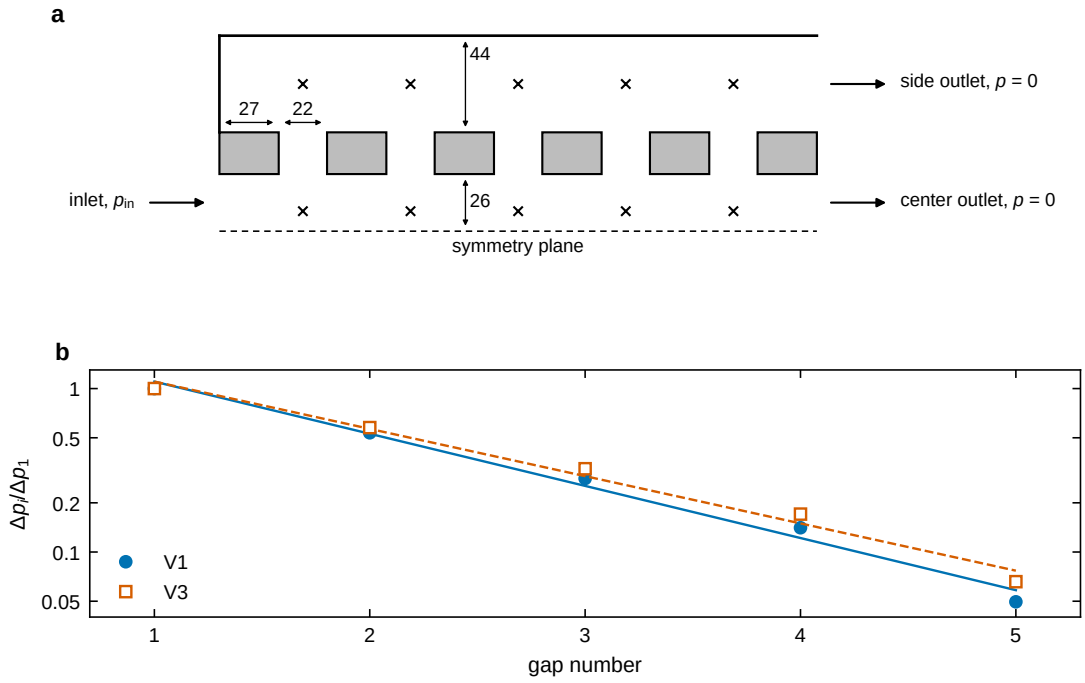
F4-3


Targets D and F, process limits, chip and parts
1
Specification
spec.py
exit 6, SPEC ERROR
2
Plan
physics/schedule.py, physics/kernels.py
3
Drawing in memory
geometry/layout.py, geometry/shapes.py
exit 5, REFUSED
4
Photomask
geometry/dxf.py
5
Measurement
checks/measure.py
6
Assessment
build.py, pipeline.assess, checks/drc.py
7
Physics audit
checks/auditor.py, checks/ruler.py
8
Design reports
checks/design_code.py
9
Verdict and record
pipeline.py, figures/proof.py
exit 0, PASS exit 2, FAIL
exit 3, outside the envelope
exit 4, INCOMPLETE

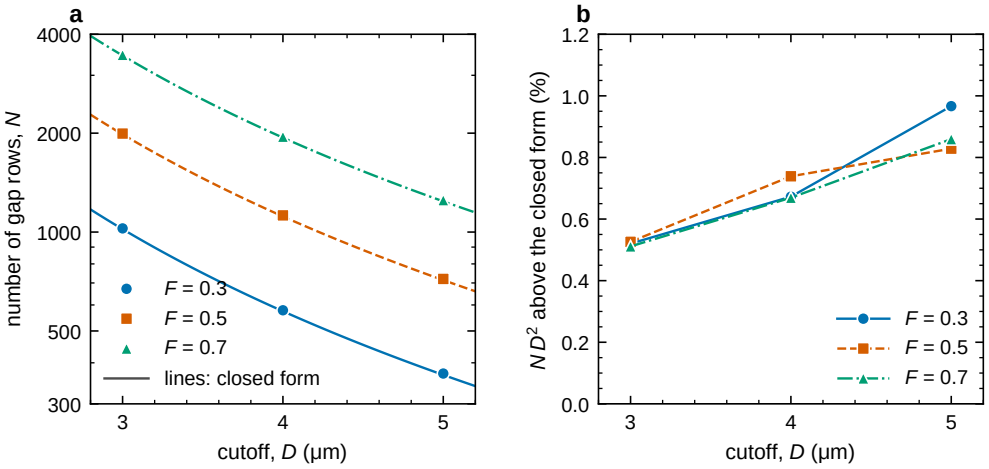
F4-4


a
b
c
run C, α = 0.80
run B, α = 1.40
run A, α = 2.69
skim width, δskim (μm)
δskim / parabolic width
δskim / exact duct law
per-gap filtration fraction, fgap (×10−4)
fgap (×10−4)
fgap (×10−4)

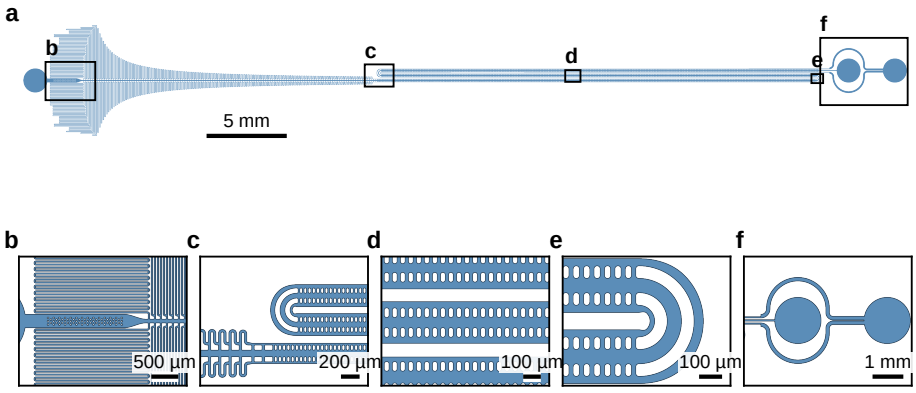
F4-5


a
44
27
22
26
side outlet, p = 0
inlet, p_in
center outlet, p = 0
symmetry plane
b
1
0.5
0.2
0.1
0.05
Δp_i/Δp_1
V1
V3
1
2
3
4
5
gap number

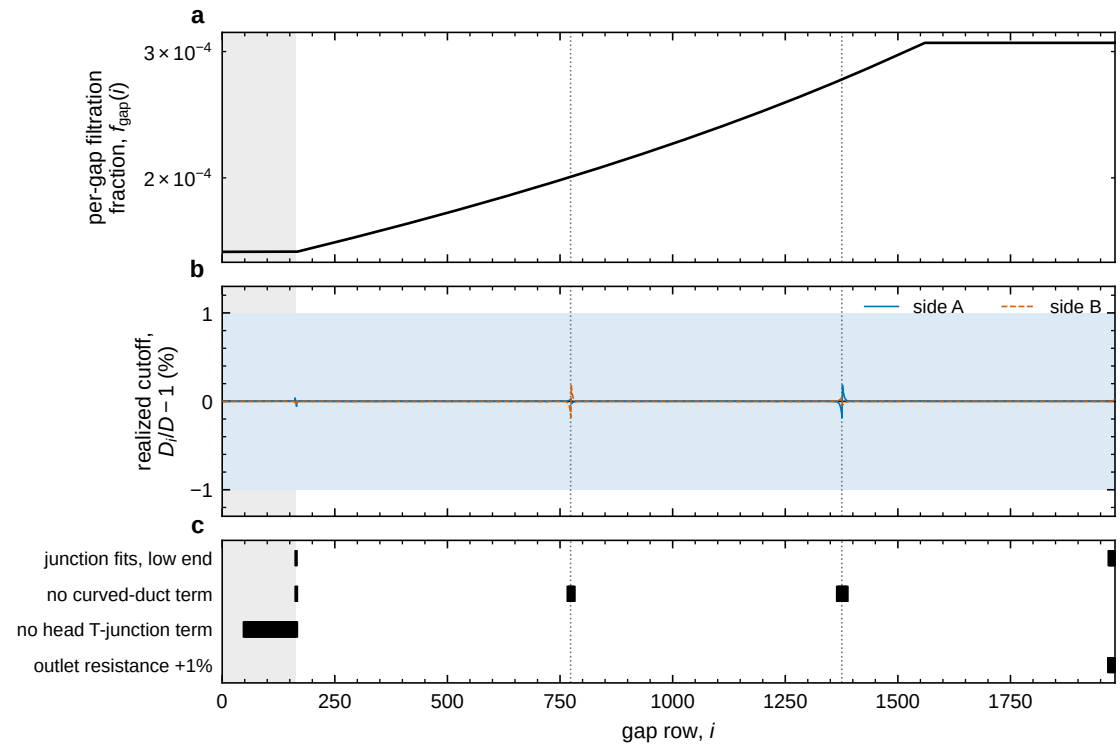
F5-1


a
4000
2000
1000
500
300
number of gap rows, N
F = 0.3
F = 0.5
F = 0.7
lines: closed form
3
4
5
cutoff, D (μm)
b
1.2
1.0
0.8
0.6
0.4
0.2
0.0
N D² above the closed form (%)
F = 0.3
F = 0.5
F = 0.7
3
4
5
cutoff, D (μm)

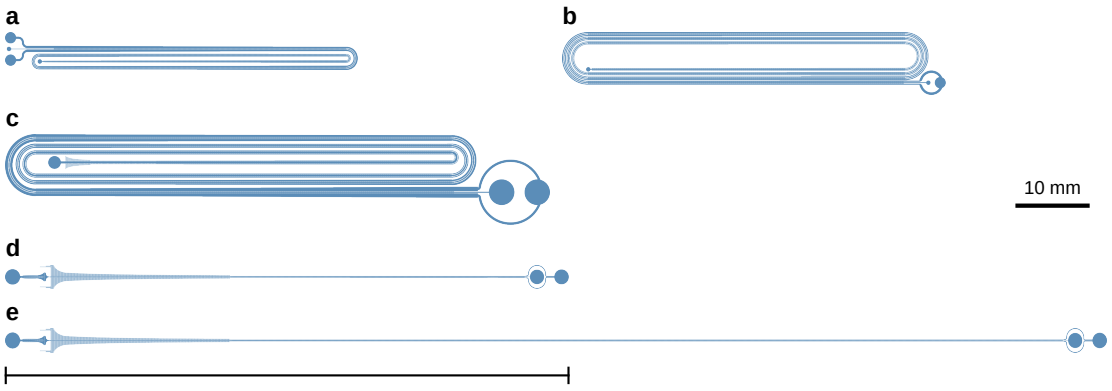
F5-2



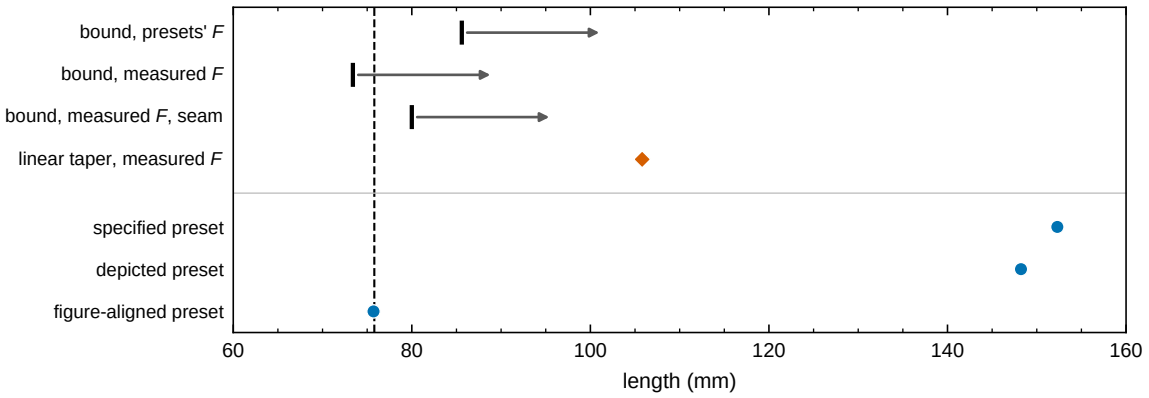
F5-3


a
b
c
per-gap filtration fraction, $f_{gap}(i)$
3×10^{-4}
2×10^{-4}
realized cutoff, $D_i/D - 1$ (%)
1
0
−1
side A
side B
junction fits, low end
no curved-duct term
no head T-junction term
outlet resistance +1%
0
250
500
750
1000
1250
1500
1750
gap row, i

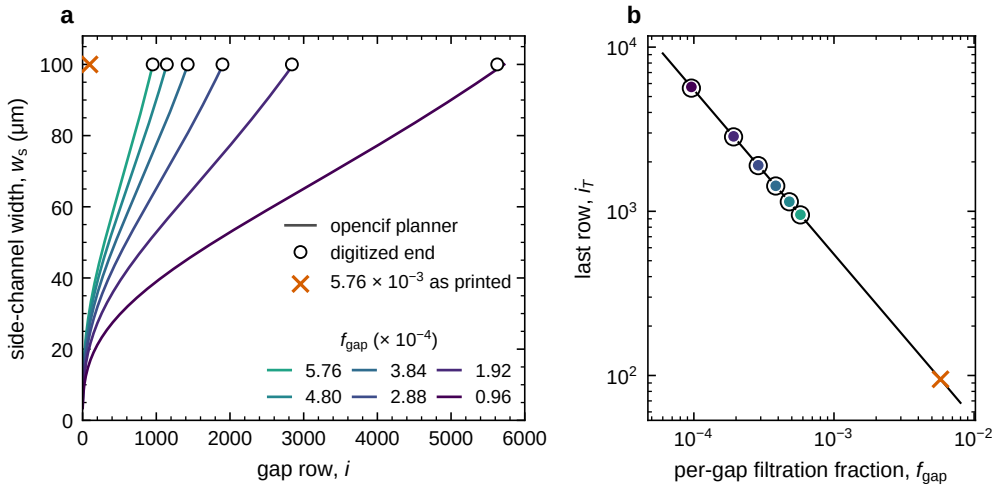
F5-4



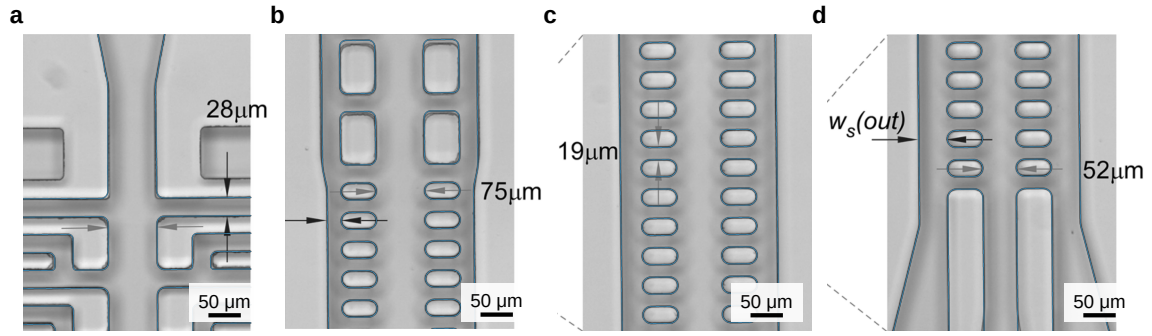
F5-5


bound, presets' F
bound, measured F
bound, measured F, seam
linear taper, measured F
specified preset
depicted preset
figure-aligned preset
60
80
100
120
140
160
length (mm)

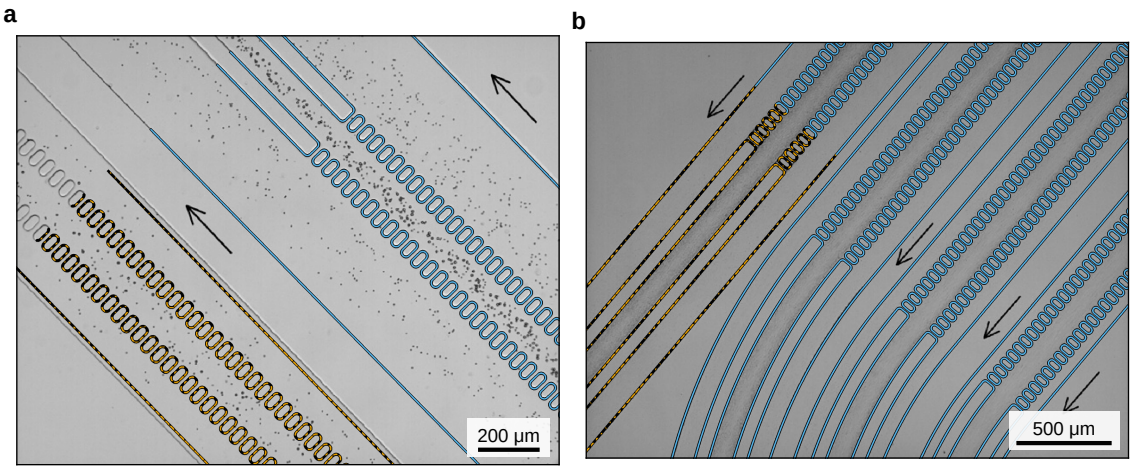
F5-6


a
b
side-channel width, w_s (µm)
gap row, i
last row, i_T
per-gap filtration fraction, f_{gap}
opencif planner
digitized end
5.76 × 10^{-3} as printed
f_{gap} (× 10^{-4})
5.76
4.80
3.84
2.88
1.92
0.96

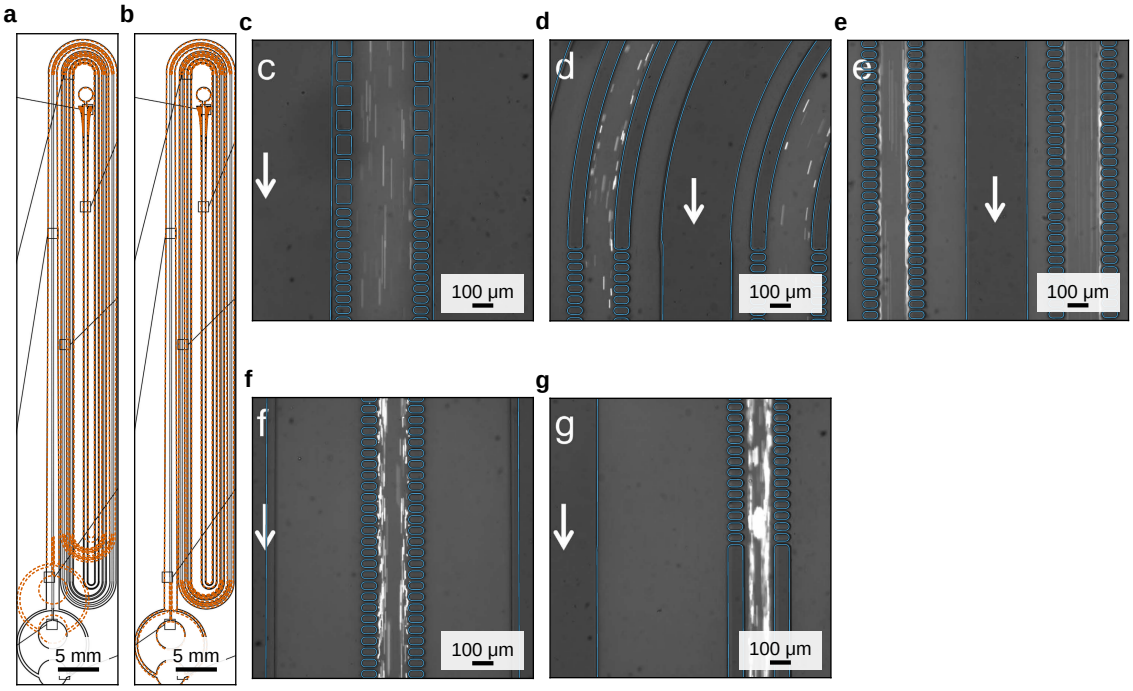
F5-7



F5-8



F5-9



F5-10